



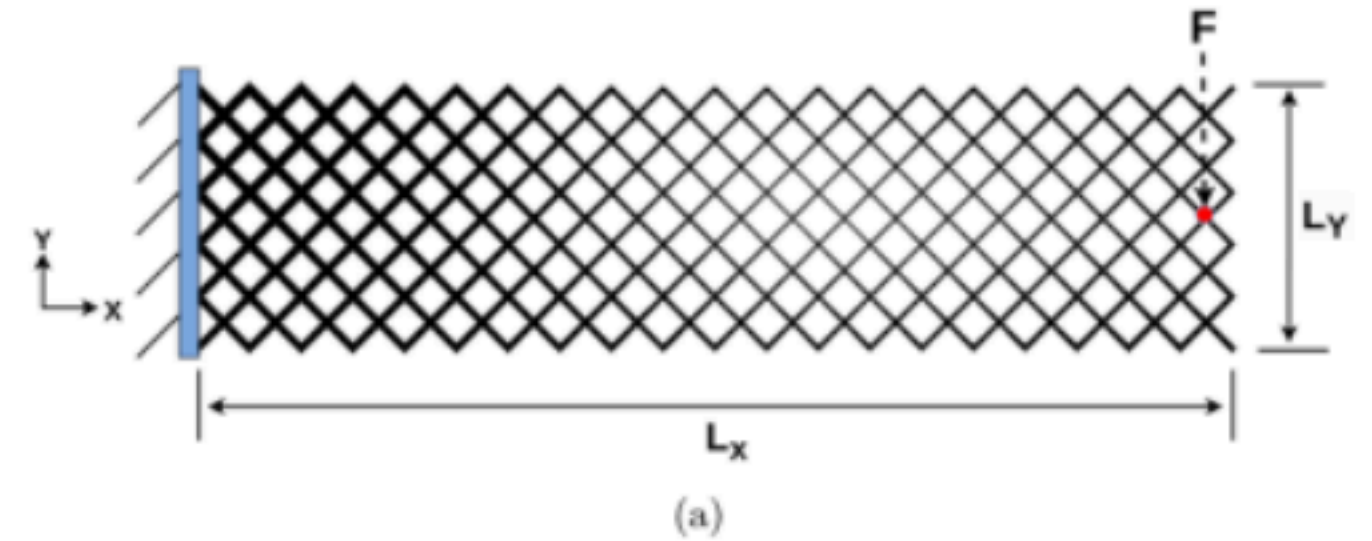
ME-504 Project

Aeshan Jyot Singh 2024MEB1326

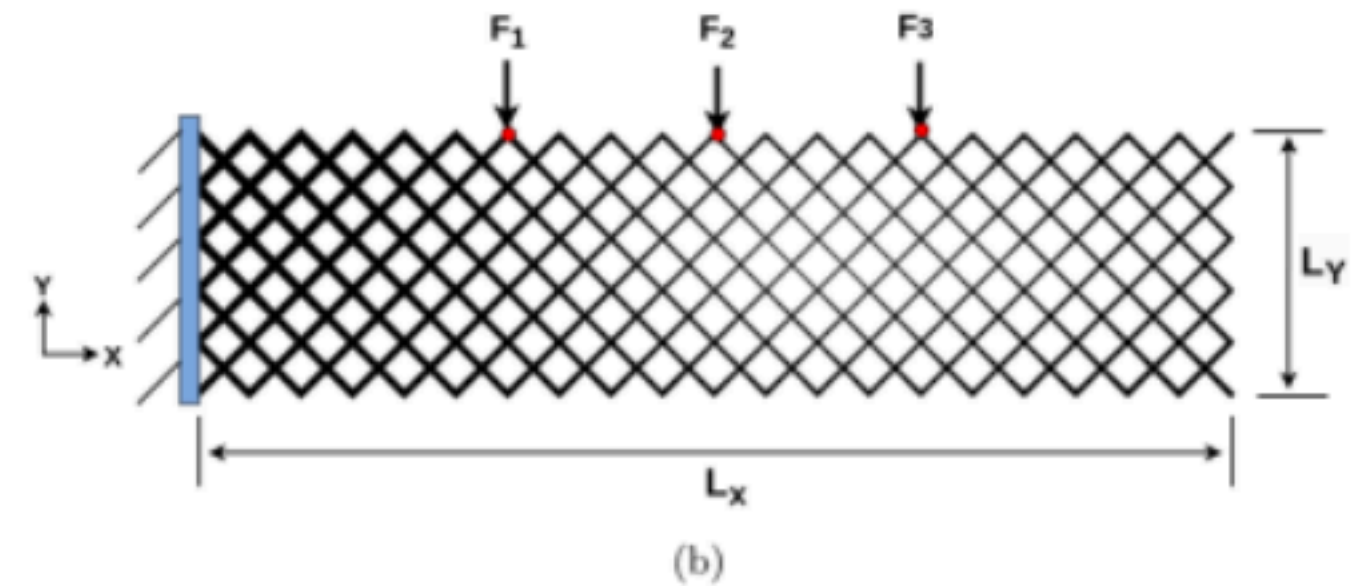
Aditya Gaurav 2024MEB1325

Cantilever Beam Configuration

- The structure consists of centered rectangular lattice unit cells forming a cantilever beam.
- The left boundary of the beam is fully fixed, restricting both horizontal and vertical displacements.



- Lattice design achieves high stiffness at low material volume (weight efficient).
- Spatial variation in nodal strut thickness alters the stiffness distribution and affects the resulting Von Mises stress field.

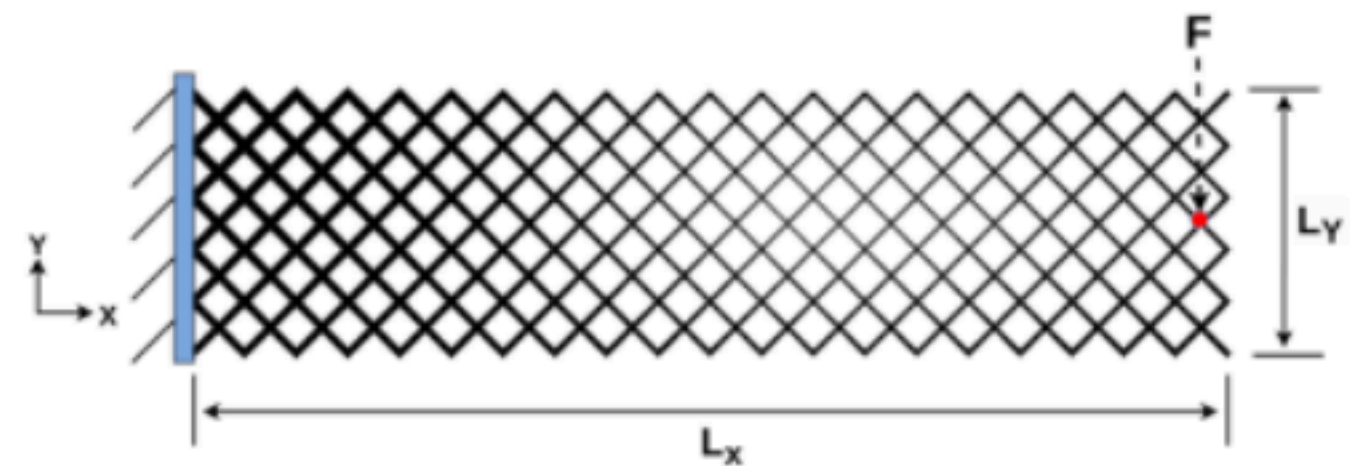


- Applied transverse loading produces bending deformation in the cantilever lattice beam
- Bending moment increases toward the fixed support

$$M = F \times (x)$$

- Upper lattice members experience compressive stress, while lower members undergo tensile stress

- Maximum Von Mises stress develops near the clamped boundary and load application region
- Thin struts in high-stress regions exhibit reduced stiffness and become critical failure locations



- Increasing local strut thickness improves stiffness and redistributes stress more

effectively Von Mises stress σ_v evaluates the localized distortional strain energy

density.

- Strut thickness directly influences the cross-sectional area and stiffness of lattice members $A = t^2$

- Stress developed in each strut depends on the applied load and effective area $\sigma = F/A$

- Small reductions in thickness significantly increase local stress and deformation • The spatial thickness distribution governs stress concentration zones and potential failure regions within the cantilever lattice beam

Von Mises Stress Criterion

- Lattice members experience combined normal and shear stresses during bending deformation.

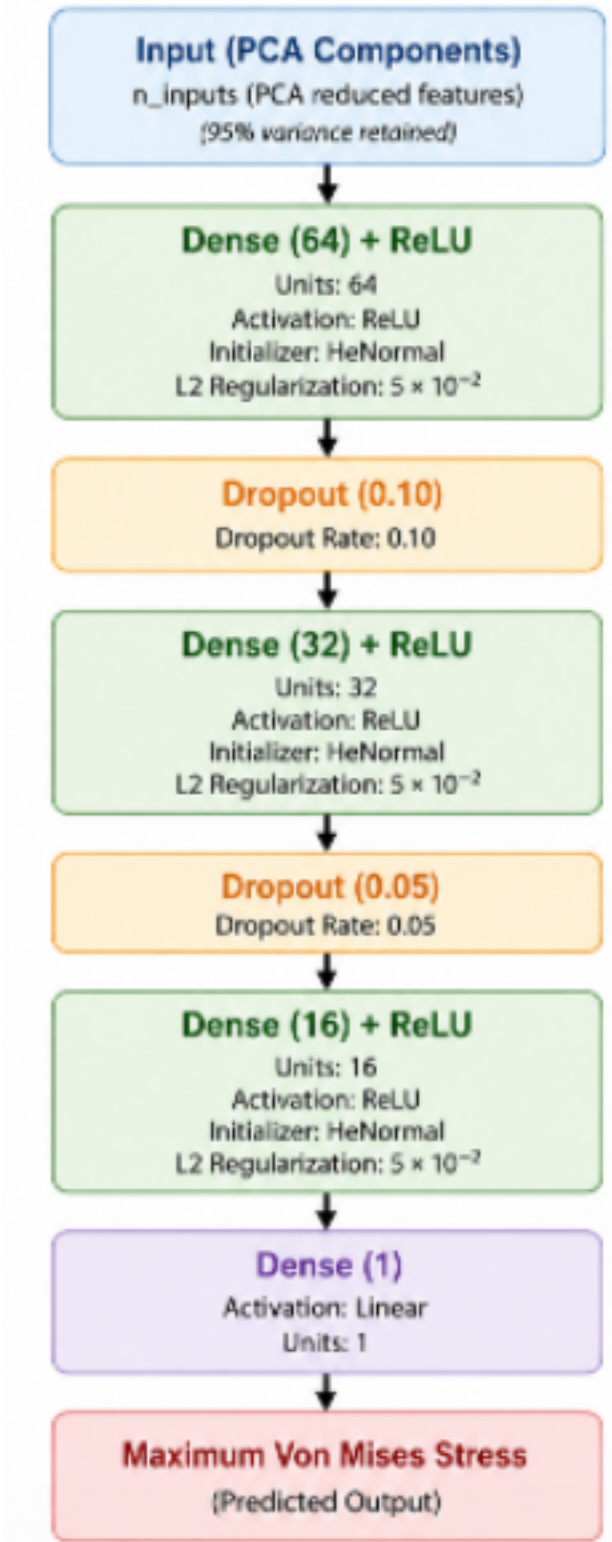
- Structural failure cannot be determined using a single stress component alone, since axial and shear effects act simultaneously within the struts.
- The Von Mises criterion combines all stress components into an equivalent stress measure used for yielding prediction:
- Yielding begins when the equivalent Von Mises stress exceeds the material yield strength.
- Local strut thickness controls the effective cross-sectional area and stiffness of the lattice members.
 $\sigma = F/A$, $A \propto t^2$
- Thin struts develop higher stress concentration and become critical failure regions within the cantilever beam.

Develop a deep neural-network model to predict the maximum Von Mises stress in centered rectangular latticecantilever beams directly from the nodal strut-thickness distribution

Model: "sequential"

Layer (type)	Output Shape	Param #
Hidden1 (Dense)	(None, 64)	13,440
Relu1 (Activation)	(None, 64)	0
Drop1 (Dropout)	(None, 64)	0
Hidden2 (Dense)	(None, 32)	2,080
Relu2 (Activation)	(None, 32)	0
Drop2 (Dropout)	(None, 32)	0
Hidden3 (Dense)	(None, 16)	528
Relu3 (Activation)	(None, 16)	0
Output (Dense)	(None, 1)	17

Total params: 16,065 (62.75 KB)
Trainable params: 16,065 (62.75 KB)
Non-trainable params: 0 (0.00 B)



Model used

Total Samples : 5000

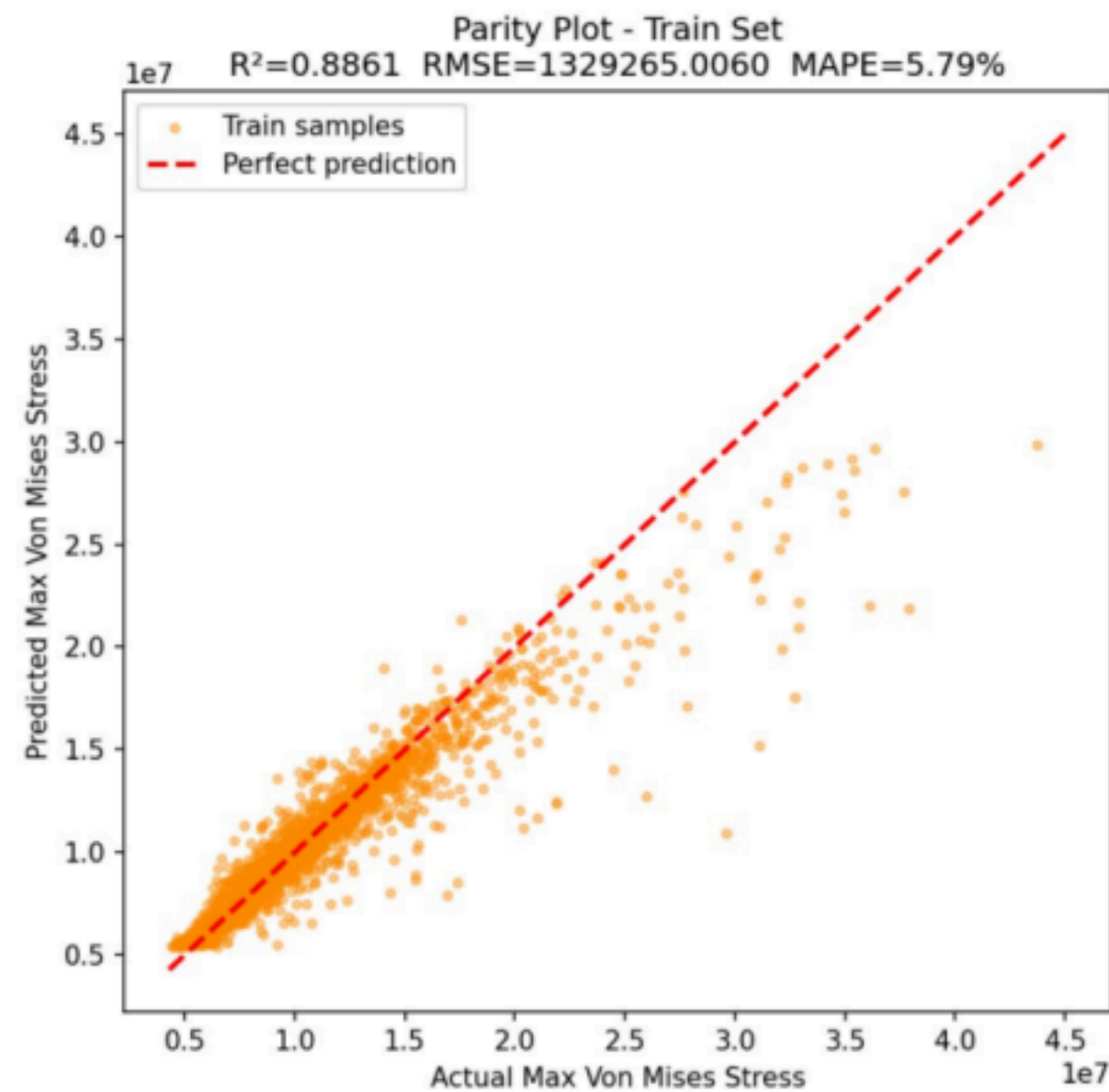
PCA Variance Retained : 95%

Train-Test Split : 80% / 20%

Batch Size : 64

Maximum Epochs : 200

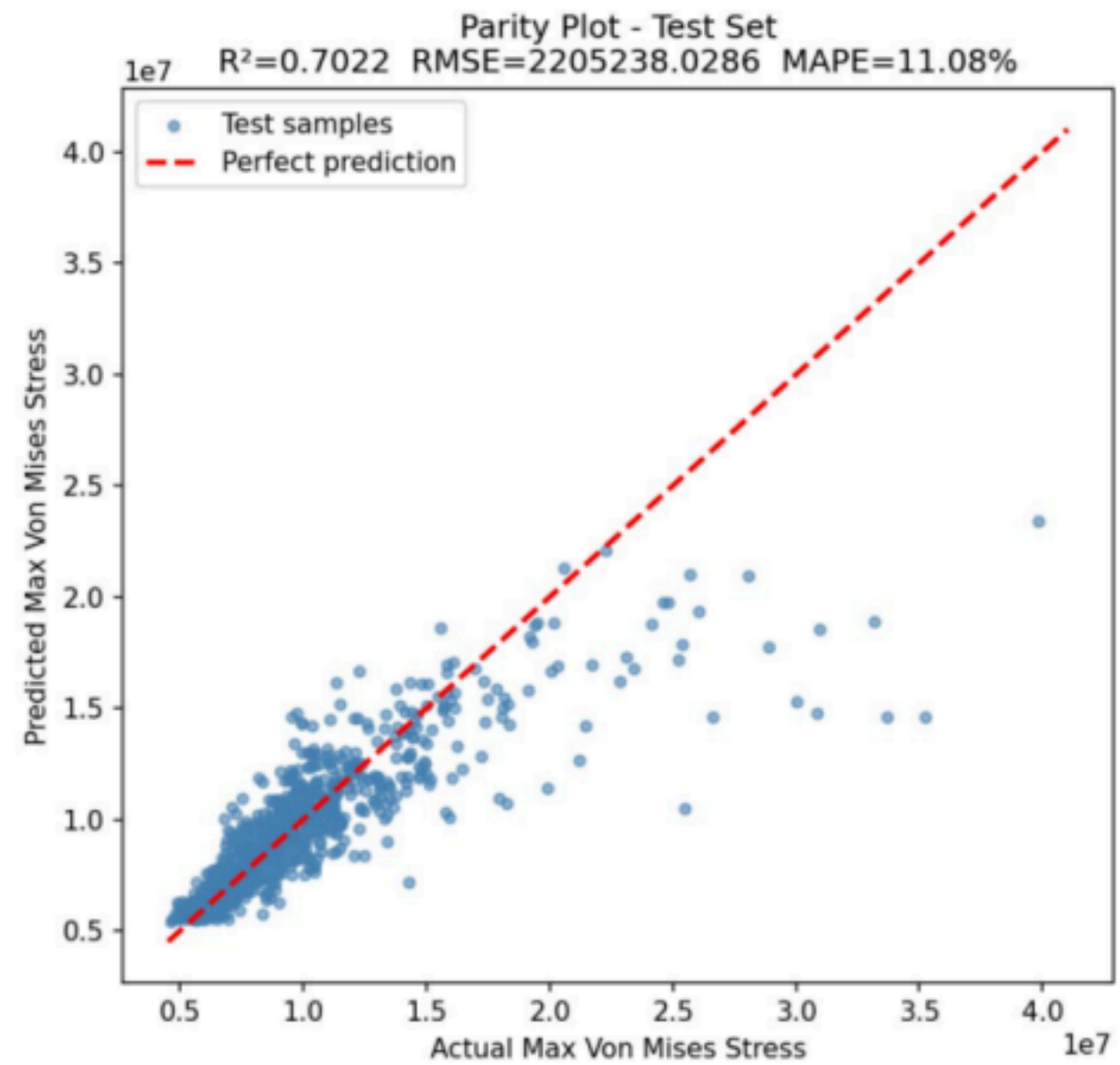
Plots:



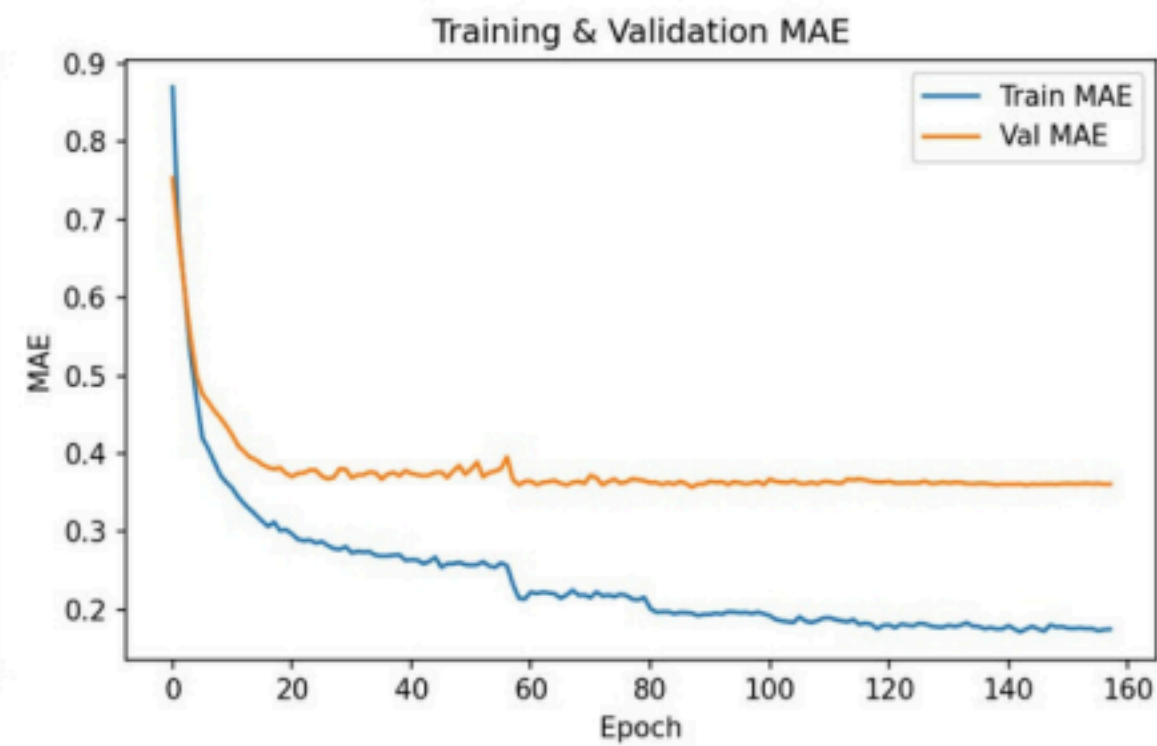
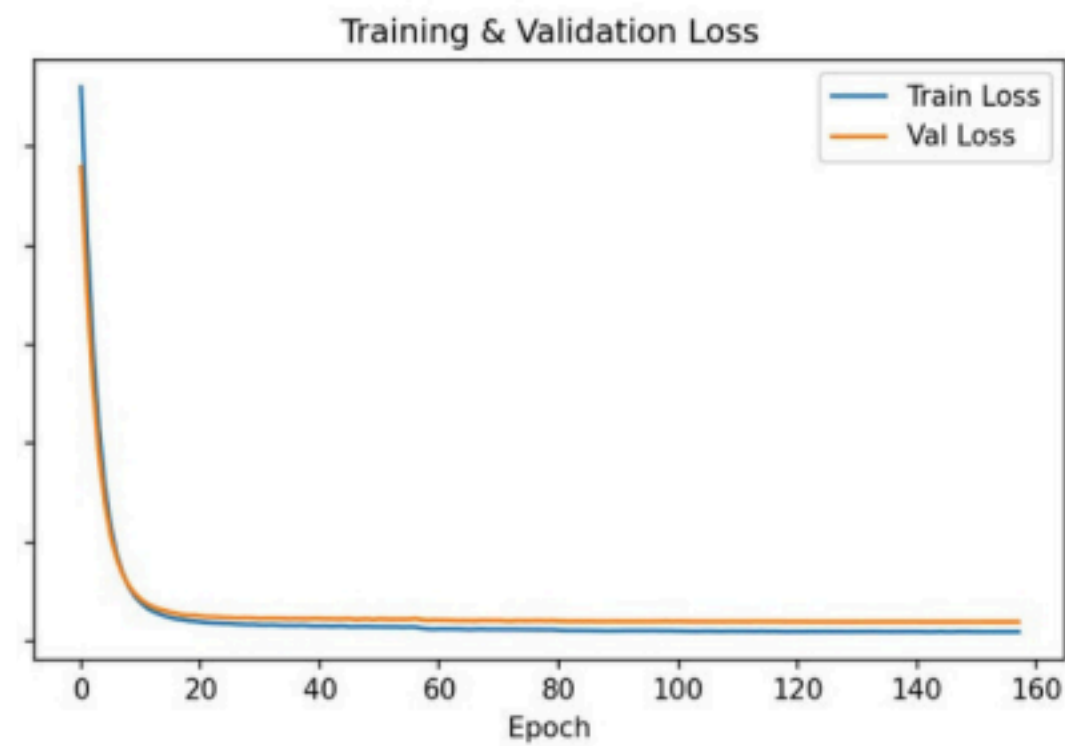
Validation Split : 20% Learning Rate : 0.001 Dropout Rate 1 : 0.10 Dropout Rate 2 : 0.05 L2 Regularization : 5×10^{-2}

Loss Function : Mean Squared Error (MSE)

Evaluation Metrics : MAE, RMSE, R², MAPE



Plots:



Results:

```

===== TEST SET METRICS =====
R²   : 0.70236
MAE   : 1213024.75000
RMSE  : 2204740.13086
MAPE  : 11.241 %

===== TRAIN SET METRICS =====
R²   : 0.88471
MAE   : 610530.31250
RMSE  : 1337212.07313
MAPE  : 5.570 %

```

Develop a neural network model that takes input the deformation values a thickness distribution and ouputs the

stressvalues at that given coordinate for that distribution.The model is only trained for the specific distribution. Total

Model used

Layer (type)	Output Shape	Param #	Connected to
Geometry_Input (InputLayer)	(None, 226)	0	-
dense (Dense)	(None, 128)	29,056	Geometry_Input[0][0]
Spatial_Input (InputLayer)	(None, 600, 2)	0	-
dense_1 (Dense)	(None, 64)	8,256	dense[0][0]
Positional_Encoding_Layer (Lambda)	(None, 600, 22)	0	Spatial_Input[0][0]
Geo_Embedding (Dense)	(None, 32)	2,080	dense_1[0][0]
dense_2 (Dense)	(None, 600, 64)	1,472	Positional_Encoding_Layer..
reshape (Reshape)	(None, 1, 32)	0	Geo_Embedding[0][0]
dense_3 (Dense)	(None, 600, 32)	2,080	dense_2[0][0]
tile (Tile)	(None, 600, 32)	0	reshape[0][0]
Spatial_Embedding (Dense)	(None, 600, 32)	1,056	dense_3[0][0]
concatenate_1 (Concatenate)	(None, 600, 64)	0	tile[0][0], Spatial_Embedding[0][0]
dense_4 (Dense)	(None, 600, 64)	4,160	concatenate_1[0][0]
dense_5 (Dense)	(None, 600, 32)	2,080	dense_4[0][0]
dense_6 (Dense)	(None, 600, 16)	528	dense_5[0][0]
Stress_Output (Dense)	(None, 600, 1)	17	dense_6[0][0]
flatten (Flatten)	(None, 600)	0	Stress_Output[0][0]
Total params: 50,785 (198.38 KB) Trainable params: 50,785 (198.38 KB) Non-trainable params: 0 (0.00 B)			

Original Input Features : 226 thickness parameters Spatial Coordinates
Used: X-coordinate, Y-coordinate Downsampled Spatial Nodes: 600
Train-Test Spli : 80% / 20%

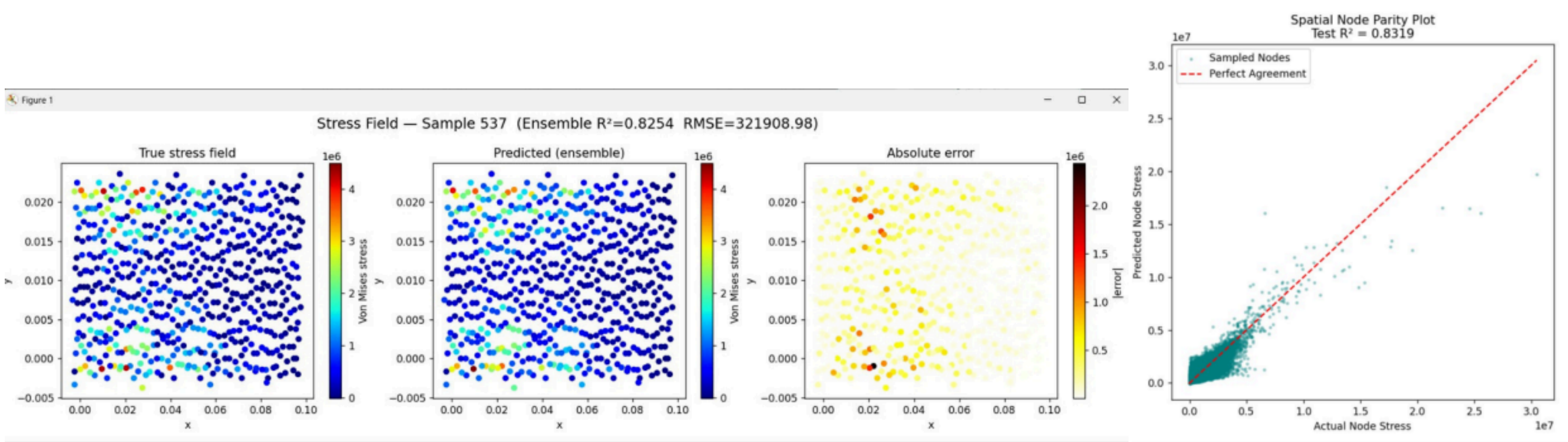
Learning Rate: 5×10^{-4}
Loss Function: Mean Squared Error (MSE) Evaluation Metrics : MAE,

Geometry Samples: 5000
RMSE, R²

Batch Size : 32
Maximum Epochs: 150
Validation Split : 20%

Early Stopping Patience: 15 epochs
Reduce LR Patience: 6 epochs
Learning Rate Reduction Factor: 0.5
Minimum Learning Rate : 1×10^{-6}

Plots:



Results:

```
===== TRAIN SET METRICS (PART 2 - OPTIMIZED) =====
Global R2 Score : 0.84004
Global MAE      : 231938.65625
Global RMSE     : 361076.75842

===== TEST SET METRICS (PART 2 - OPTIMIZED) =====
Global R2 Score : 0.83191
Global MAE      : 233918.04688
Global RMSE     : 367187.78402
```

Comparison of Complexity Between the Two Problems

Part 1: Maximum Von Mises Stress Prediction

- The first problem is computationally simpler because the neural network predicts only a single scalar output corresponding to the maximum Von Mises stress.
 - The mapping complexity is relatively low:
 - A smaller NN with three hidden layers is sufficient to learn the global stress behavior of the structure.
 - Since only one output value is predicted, the model requires:
 - fewer trainable parameters,
 - lower memory usage,
 - shorter training time,
 - and lower computational cost.
 - The data requirement is also moderate because the model only needs to learn the relationship between thickness distribution and overall maximum stress response.
- #### Part 2: Full Stress Field Prediction
- The second problem is significantly more complex because the network predicts the complete spatial stress distribution across the lattice domain.

Unlike Part 1, the model must learn stress variations at hundreds of spatial locations simultaneously while preserving spatial continuity and local stress concentration behavior.

- This requires:

a deeper and higher-capacity network,

spatial coordinate encoding,

positional embeddings,

and a two-stream architecture combining geometry and spatial information.

- The computational complexity is much higher due to:

increased output dimensionality,

larger parameter space,

and more difficult optimization.

- Part 2 also demands substantially more training data because the model must accurately learn local stress behavior throughout the entire structure instead of predicting only a single global quantity.